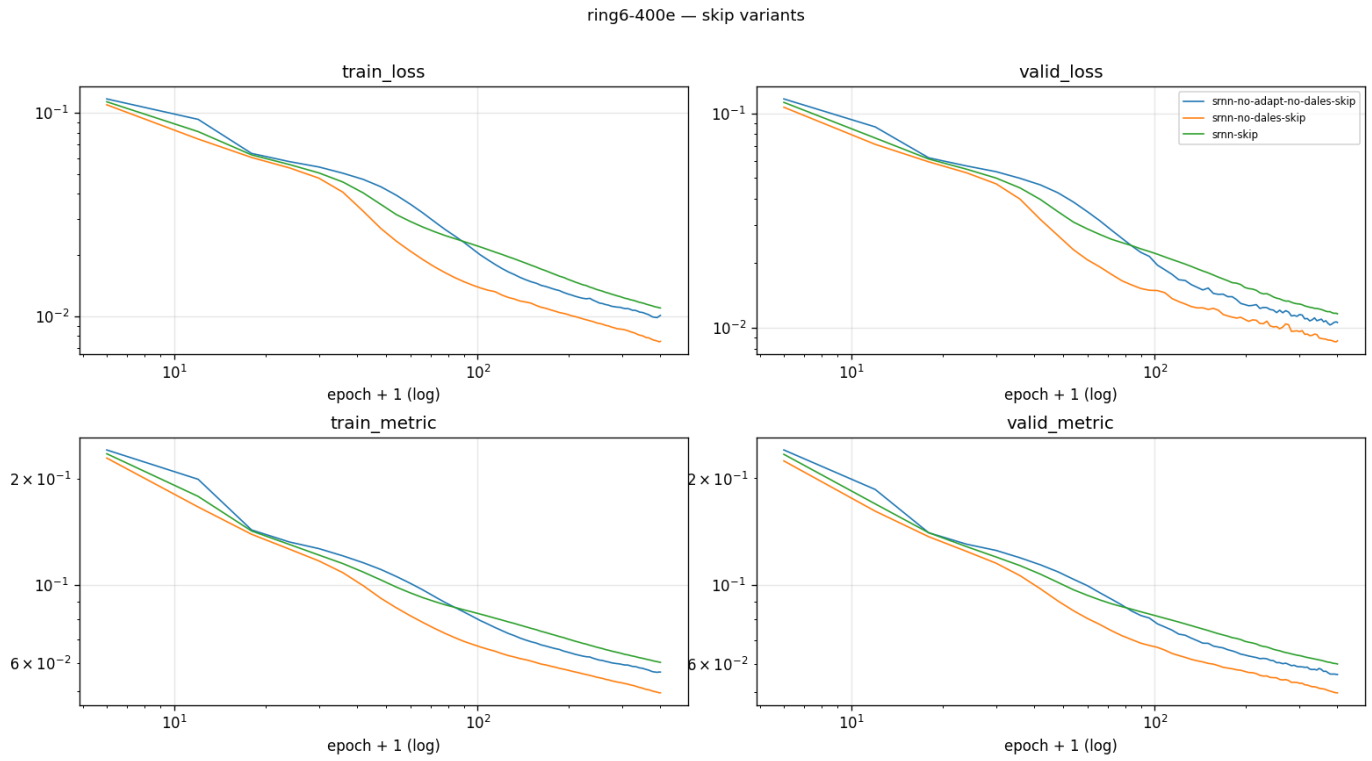


Run: ring6-400e

Log-log loss curves



Final test_loss (init - last) [skip]:
 srnn-no-adapt-no-dales-skip: 0.13692 → 0.01047
 srnn-no-dales-skip: 0.11825 → 0.00828
 srnn-skip: 0.11798 → 0.01120

Figure 1: skip variants — log-log loss/metric

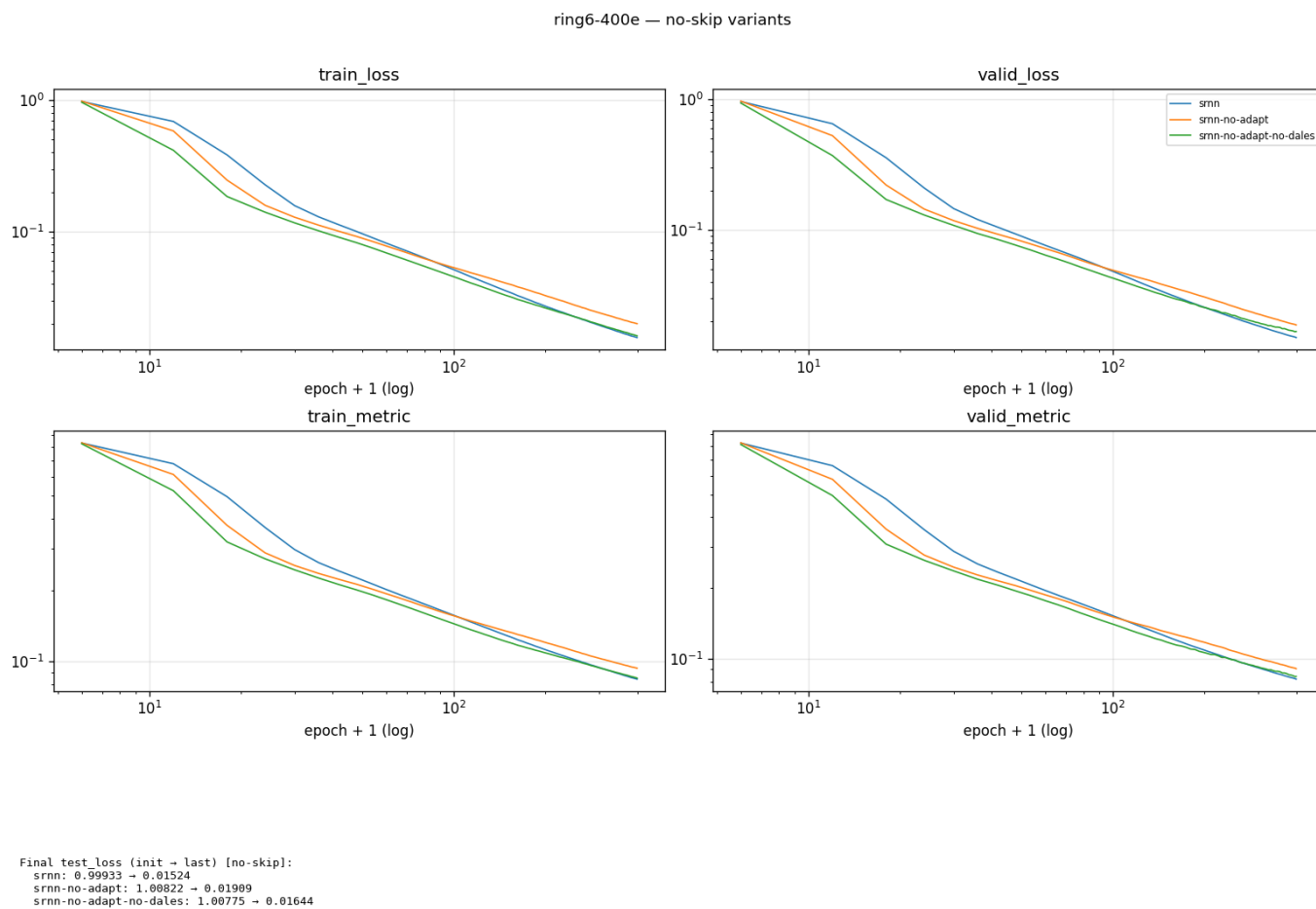


Figure 2: no-skip variants — log-log loss/metric

srnn-no-adapt-no-dales

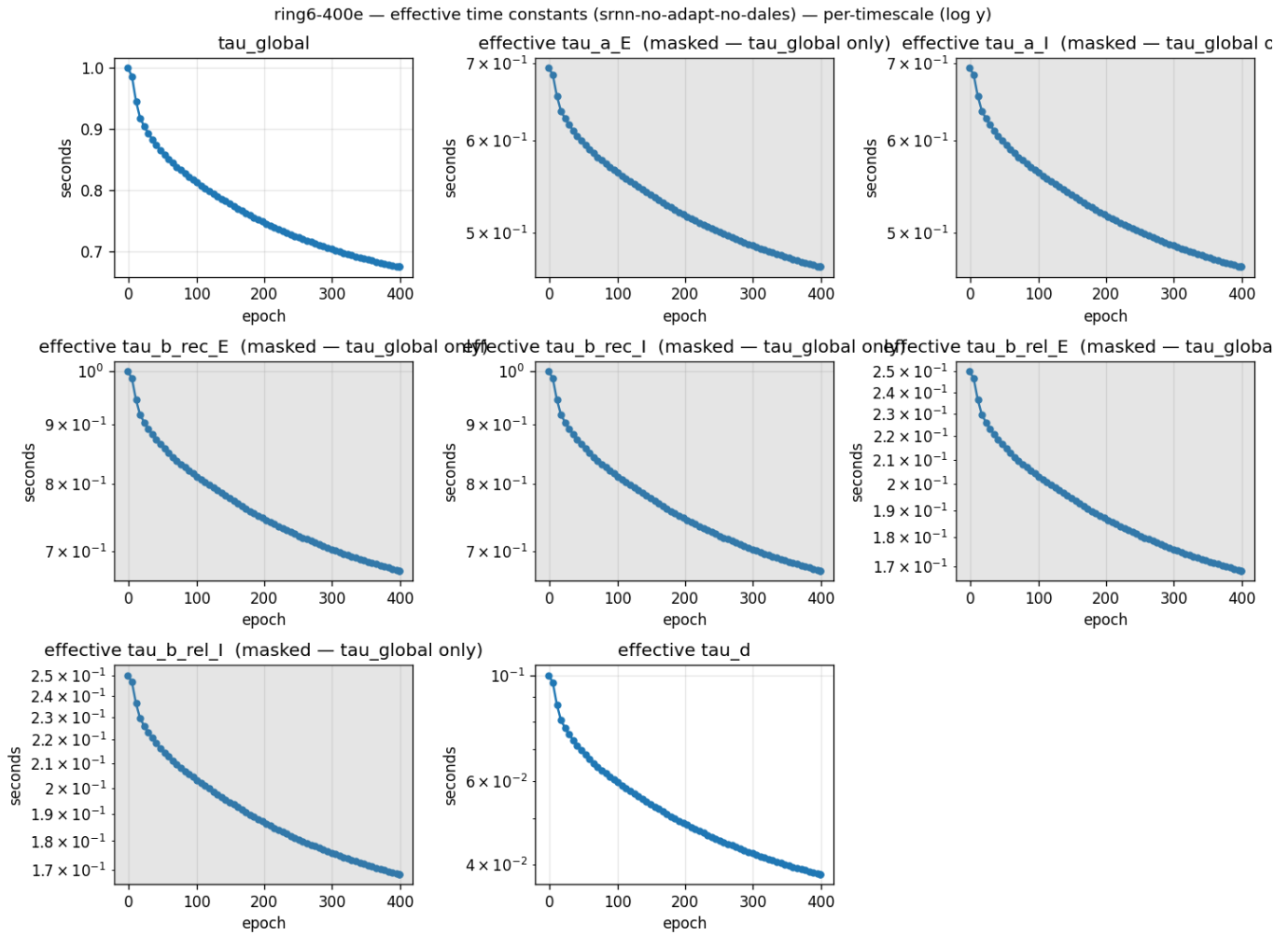
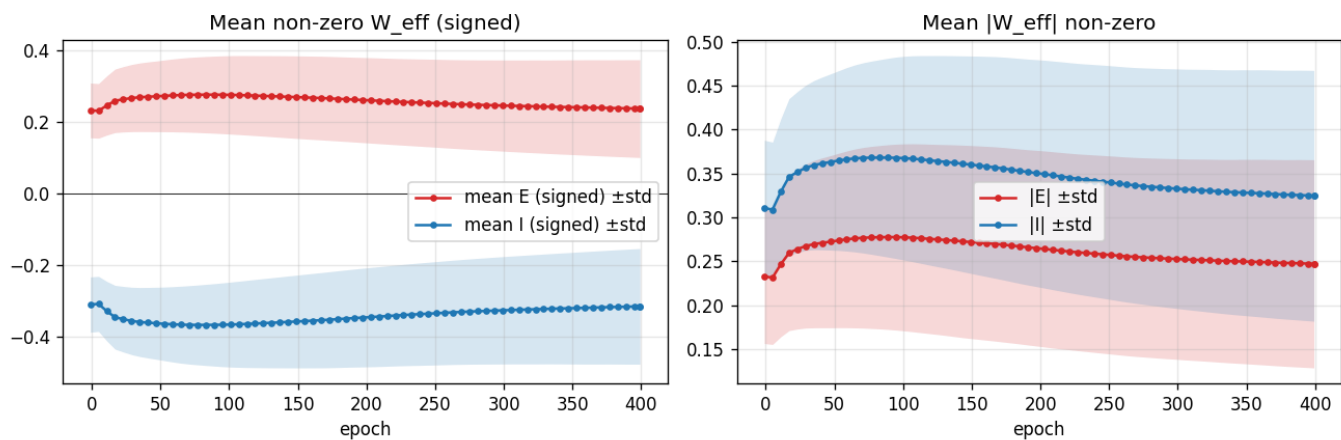
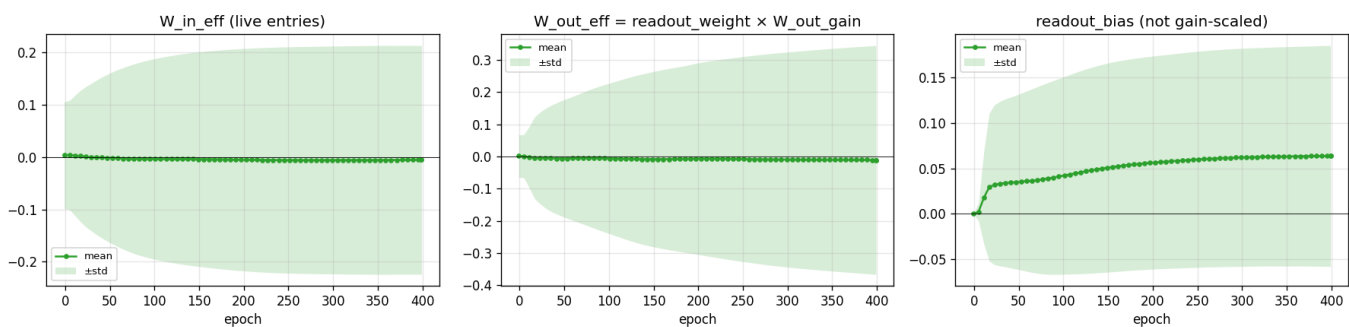


Figure 3: Tau evolution

ring6-400e — recurrent E vs I weights (srnn-no-adapt-no-dales)

Figure 4: W_{EI} evolution

ring6-400e — input / output weights (srnn-no-adapt-no-dales)

Figure 5: W_{io} evolution

— offsets and threshold (srnn-no-adapt-no-dales) — scalar-driven

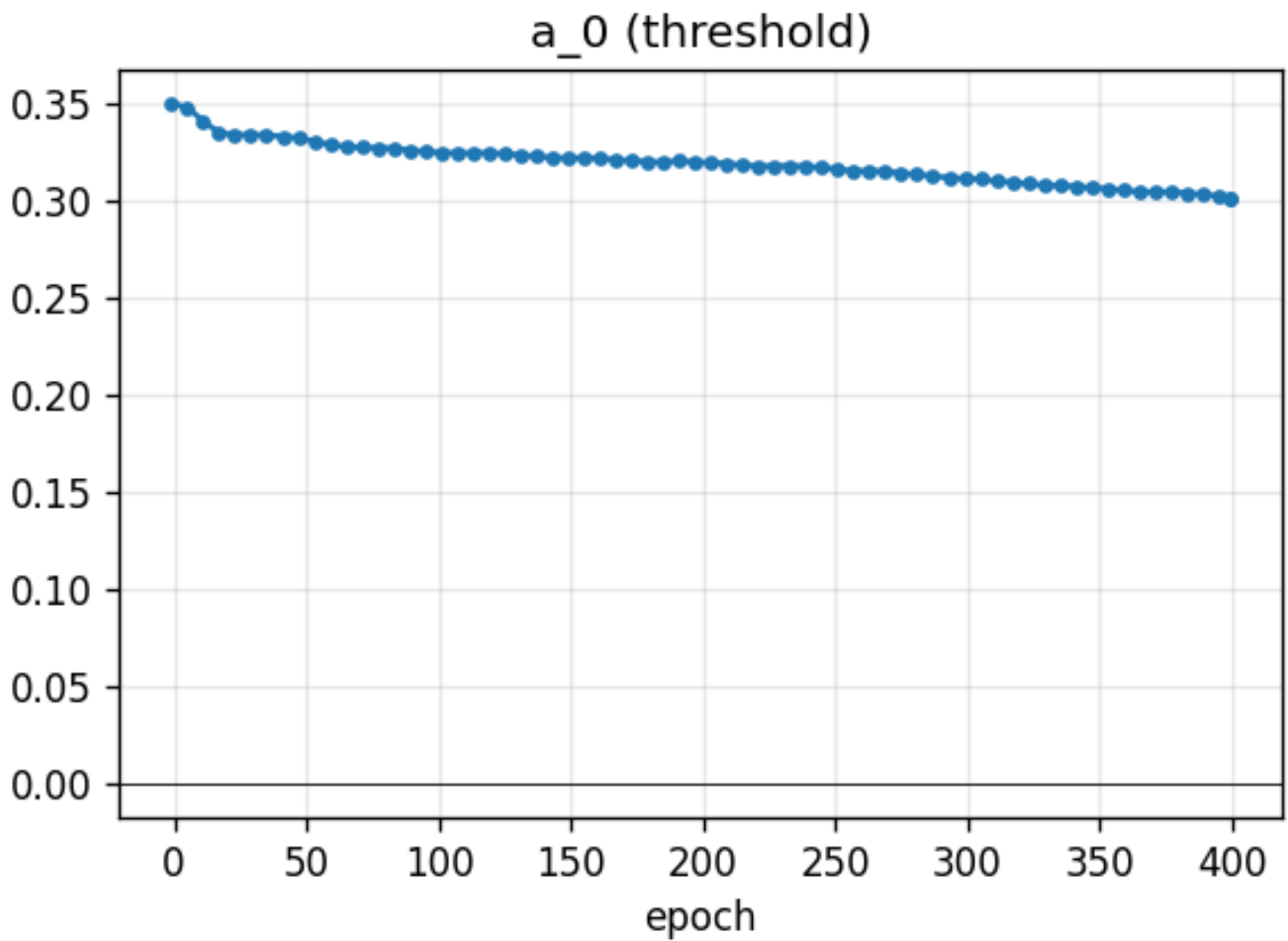


Figure 6: Offsets and threshold

srnn-no-adapt-no-dales — parameter table

```
# ring6-400e - variant srnn-no-adapt-no-dales (k=0)
# Effective values (post-transform). Means/stds computed over the indicated dimension.
```

effective parameter	init mean	final mean	init std	final std	n

tau_global (s)	+1	+0.67424	-	-	scalar
tau_d (s)	+0.1	+0.038251	0	0	300
tau_b_rec_E (s)	+1	+0.67424	0	0	150
tau_b_rel_E (s)	+0.25	+0.16856	0	0	150
tau_b_rec_I (s)	+1	+0.67424	0	0	150
tau_b_rel_I (s)	+0.25	+0.16856	0	0	150
W_eff (E src, signed)	+0.2326	+0.23755	0.07663	0.1367	15048
W_eff (I src, signed)	-0.31062	-0.31561	0.07727	0.1617	14977
W_eff (E src)	+0.23267	+0.2471	0.07641	0.1186	15048
W_eff (I src)	+0.31062	+0.32459	0.07724	0.1429	14977
W_in_eff (input neurons)	+0.0038159	-0.0053236	0.1019	0.2185	1275
a_0 (threshold)	+0.35	+0.3017	0	0	300
W_out_eff (readout × gain)	+0.0011253	-0.010918	0.06572	0.3555	1275
readout_bias	+0	+0.063545	0	0.1212	17
ic.ic (per-variant init)	+0.13192	+0.13192	0.5265	0.5265	1500

srnn-no-adapt

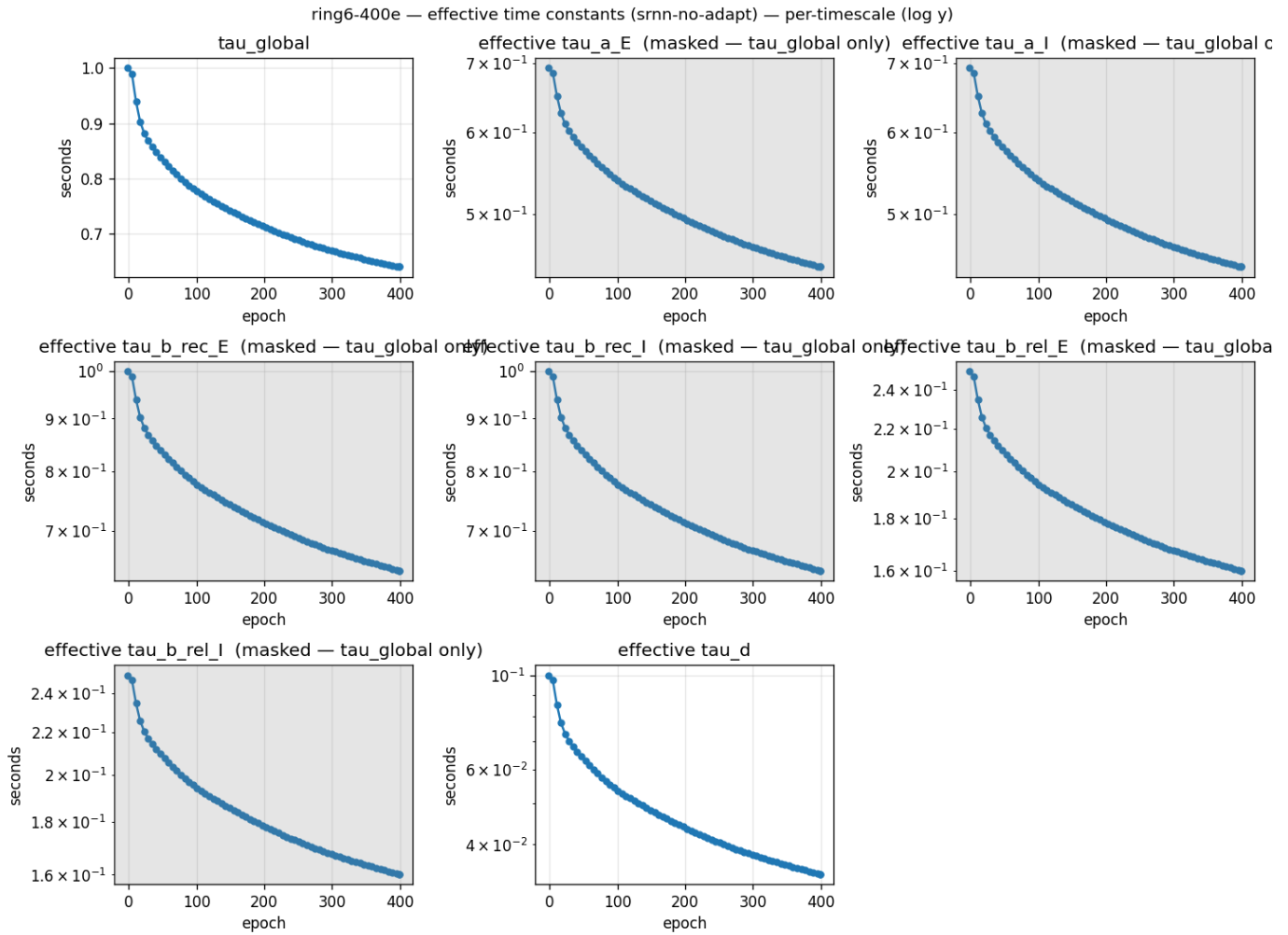
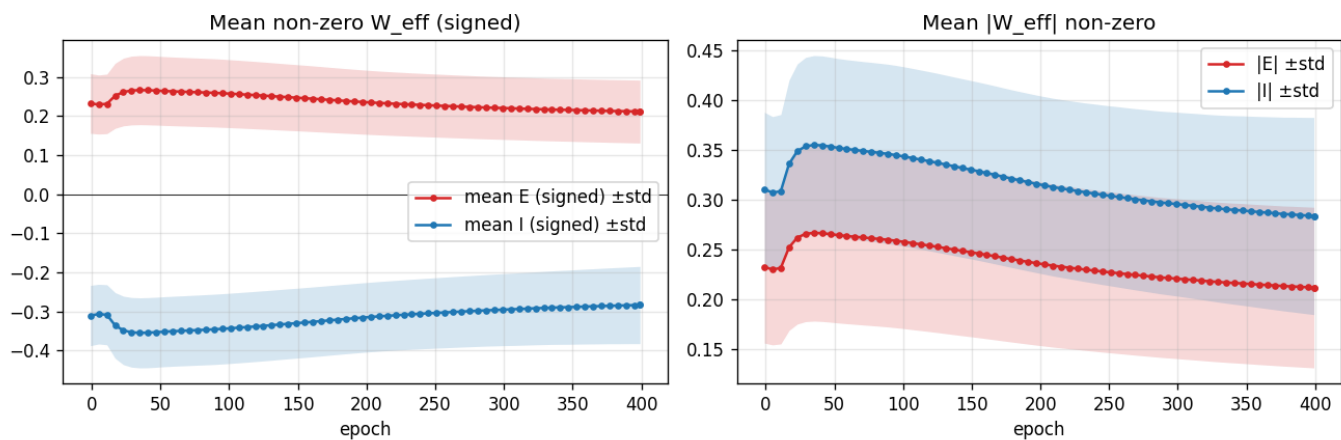
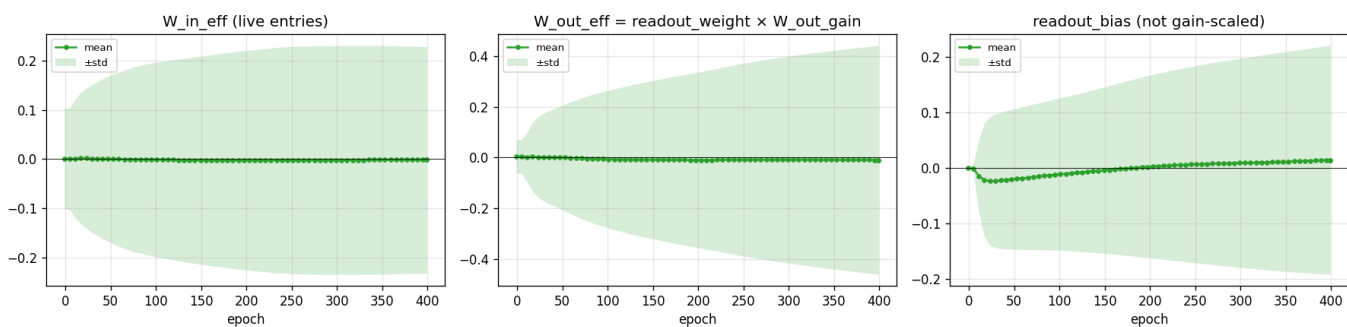


Figure 7: Tau evolution

ring6-400e — recurrent E vs I weights (srnn-no-adapt)

Figure 8: W_{EI} evolution

ring6-400e — input / output weights (srnn-no-adapt)

Figure 9: W_{io} evolution

00e — offsets and threshold (srnn-no-adapt) — scalar-driven (vec

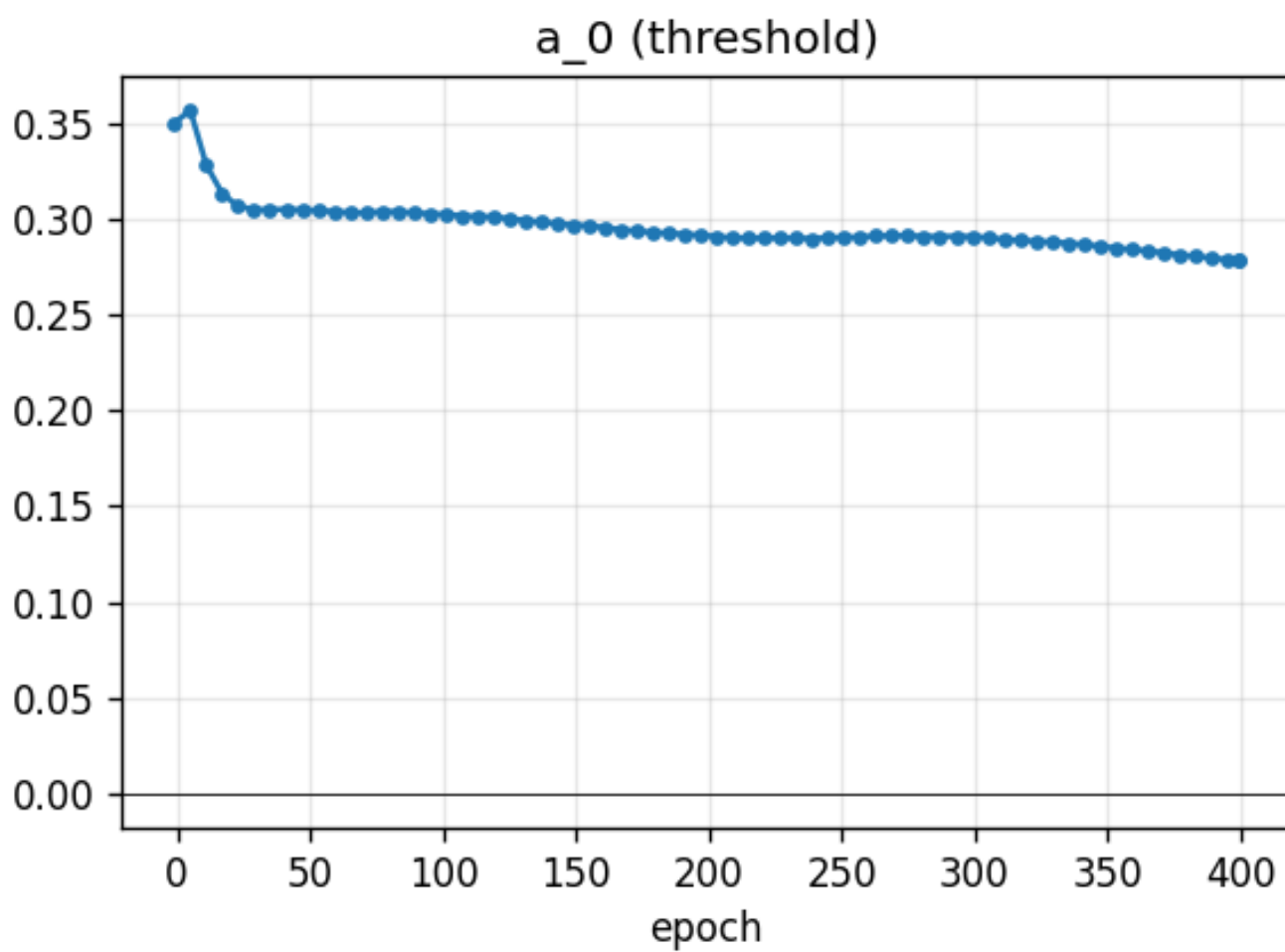


Figure 10: Offsets and threshold

srnn-no-adapt — parameter table

```
# ring6-400e - variant srnn-no-adapt (k=1)
# Effective values (post-transform). Means/stds computed over the indicated dimension.
```

effective parameter	init mean	final mean	init std	final std	n

tau_global (s)	+1	+0.64079	-	-	scalar
tau_d (s)	+0.1	+0.034076	0	0	300
tau_b_rec_E (s)	+1	+0.64079	0	0	150
tau_b_rel_E (s)	+0.25	+0.1602	0	0	150
tau_b_rec_I (s)	+1	+0.64079	0	0	150
tau_b_rel_I (s)	+0.25	+0.1602	0	0	150
W_eff (E src, signed)	+0.23267	+0.21199	0.07641	0.08059	15048
W_eff (I src, signed)	-0.31062	-0.28378	0.07724	0.09891	14977
W_eff (E src)	+0.23267	+0.21199	0.07641	0.08059	15048
W_eff (I src)	+0.31062	+0.28378	0.07724	0.09891	14977
W_in_eff (input neurons)	+0.00075891	-0.0015403	0.1012	0.2295	1275
a_0 (threshold)	+0.35	+0.27806	0	0	300
W_out_eff (readout × gain)	+0.0034456	-0.0094228	0.06605	0.4501	1275
readout_bias	+0	+0.013928	0	0.2061	17
ic.ic (per-variant init)	+0.13096	+0.13096	0.4725	0.4725	1500

srnn-no-dales-skip

ring6-400e — effective time constants (srnn-no-dales-skip) — per-timescale (log y)

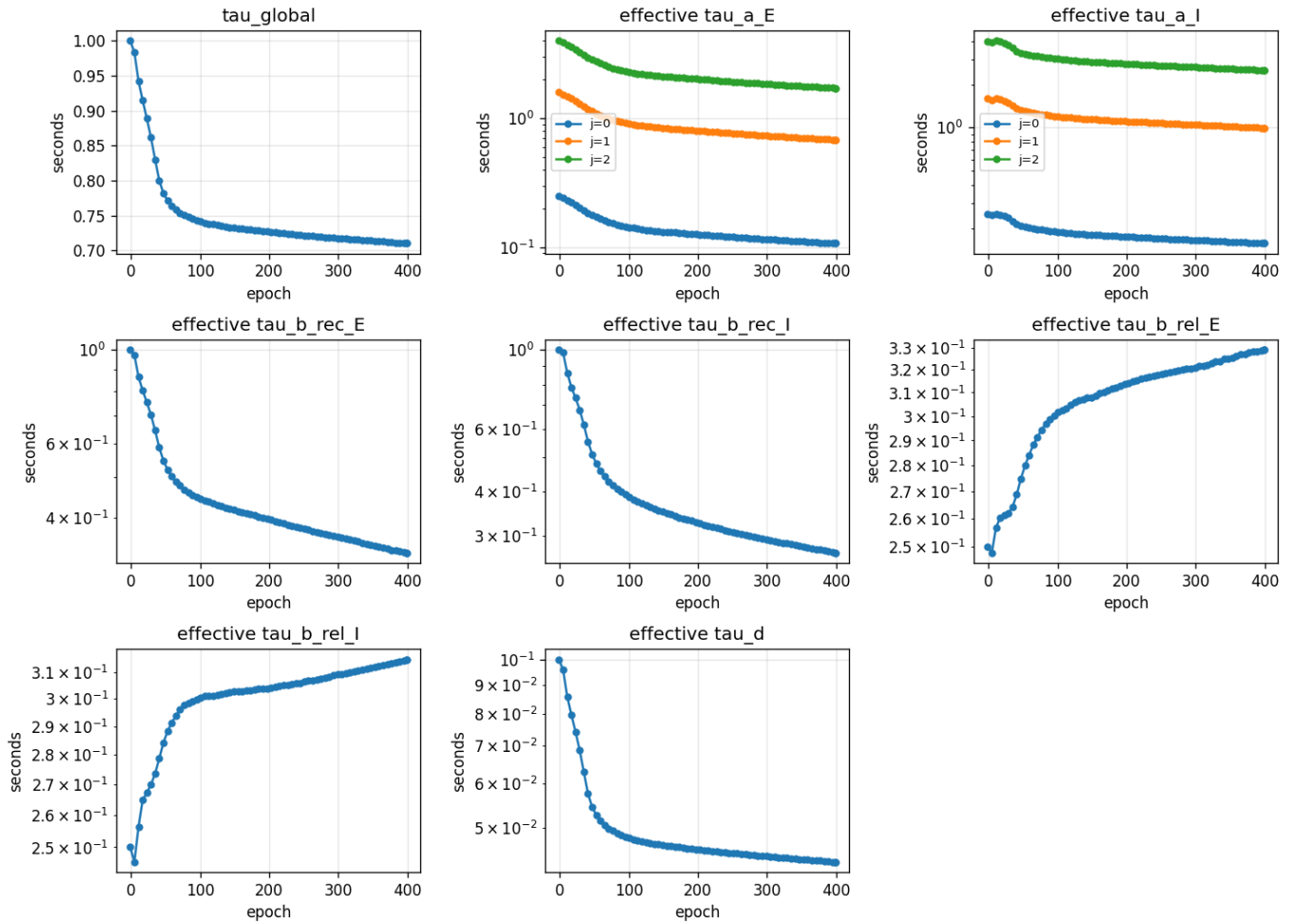


Figure 11: Tau evolution

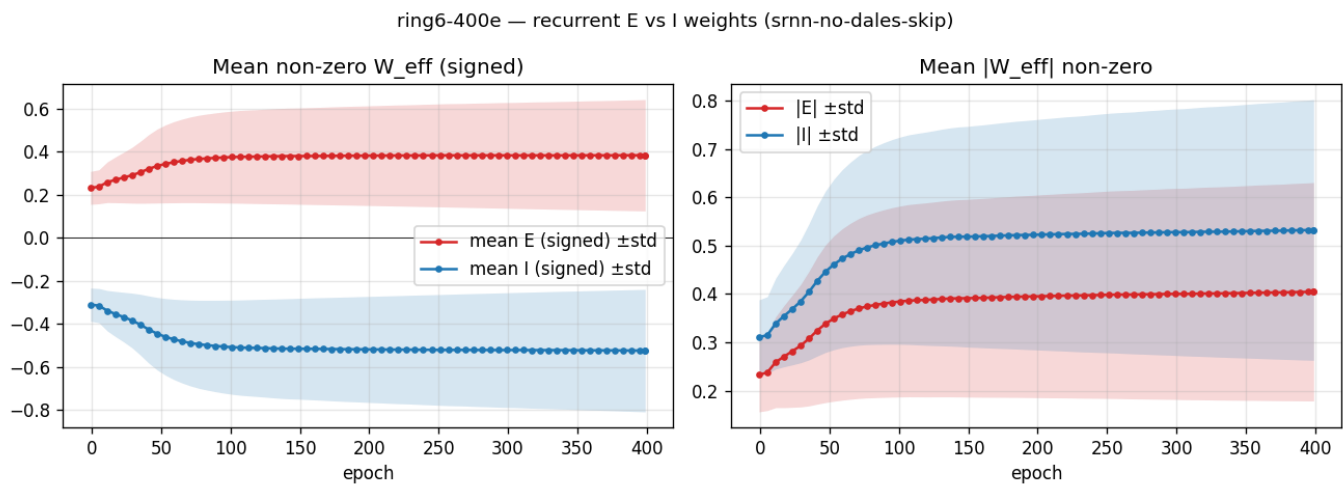
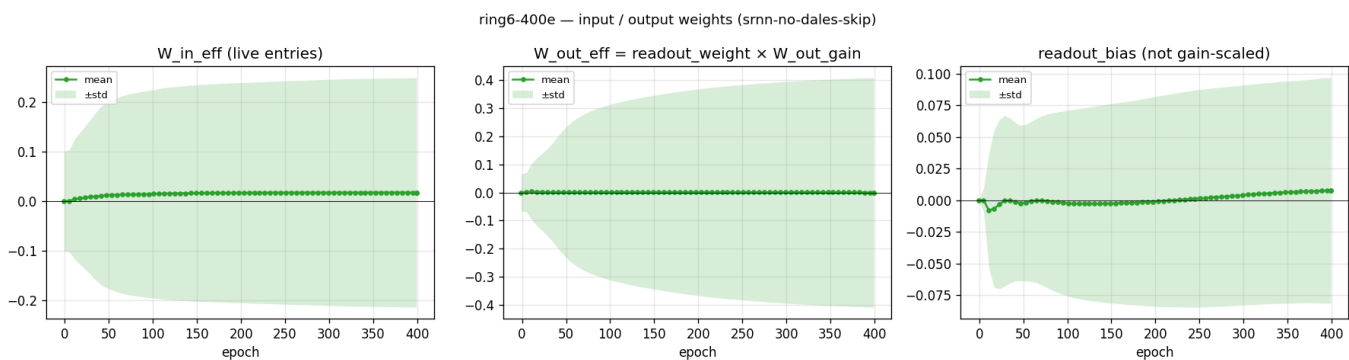
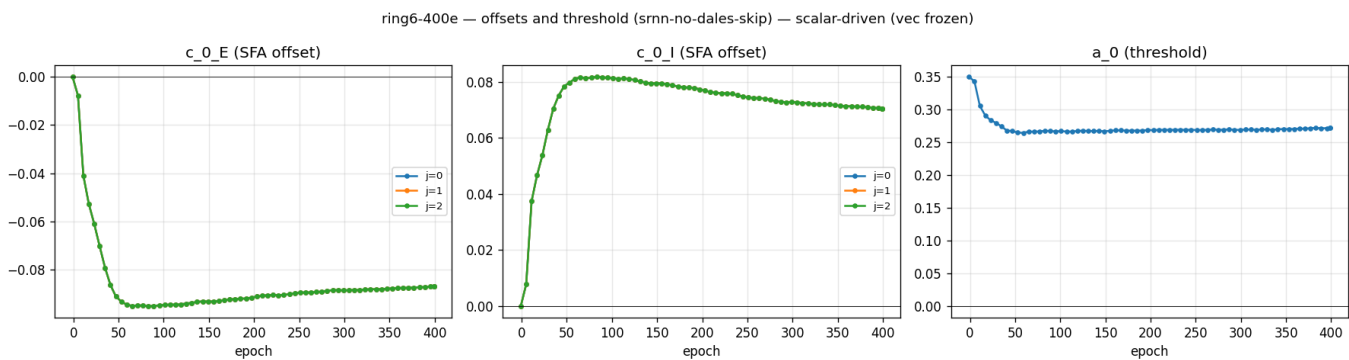
Figure 12: W_{EI} evolutionFigure 13: W_{io} evolution

Figure 14: Offsets and threshold

srnn-no-dales-skip — parameter table

ring6-400e - variant srnn-no-dales-skip (k=2)
Effective values (post-transform). Means/stds computed over the indicated dimension.

effective parameter	init mean	final mean	init std	final std	n

tau_global (s)	+1	+0.71013	-	-	scalar
tau_d (s)	+0.1	+0.043403	0	0	300
tau_a_E[j=0] (s)	+0.25	+0.10724	0	0	150
tau_a_E[j=1] (s)	+1.5896	+0.68185	0	0	150
tau_a_E[j=2] (s)	+4	+1.7158	0	0	150
tau_a_I[j=0] (s)	+0.25	+0.15645	0	0	150
tau_a_I[j=1] (s)	+1.5896	+0.99477	0	0	150
tau_a_I[j=2] (s)	+4	+2.5032	0	0	150
tau_b_rec_E (s)	+1	+0.33078	0	0	150
tau_b_rel_E (s)	+0.25	+0.32892	0	0	150
tau_b_rec_I (s)	+1	+0.26886	0	0	150
tau_b_rel_I (s)	+0.25	+0.31483	0	0	150
W_eff (E src, signed)	+0.2326	+0.38439	0.07663	0.2588	15048
W_eff (I src, signed)	-0.31062	-0.52445	0.07727	0.2847	14977
W_eff (E src)	+0.23267	+0.40465	0.07641	0.2258	15048
W_eff (I src)	+0.31062	+0.53241	0.07724	0.2695	14977
W_in_eff (input neurons)	-0.00021194	+0.017065	0.09952	0.2314	1275
c_E[j=0] (SFA coupling)	+0.05	+0.041108	0	0	150
c_E[j=1] (SFA coupling)	+0.05	+0.041108	0	0	150
c_E[j=2] (SFA coupling)	+0.05	+0.041108	0	0	150
c_0_E[j=0] (SFA offset)	+0	-0.086949	0	0	150
c_0_E[j=1] (SFA offset)	+0	-0.086949	0	0	150
c_0_E[j=2] (SFA offset)	+0	-0.086949	0	0	150
c_I[j=0] (SFA coupling)	+0.05	+0.04975	0	0	150
c_I[j=1] (SFA coupling)	+0.05	+0.04975	0	0	150
c_I[j=2] (SFA coupling)	+0.05	+0.04975	0	0	150
c_0_I[j=0] (SFA offset)	+0	+0.070548	0	0	150
c_0_I[j=1] (SFA offset)	+0	+0.070548	0	0	150
c_0_I[j=2] (SFA offset)	+0	+0.070548	0	0	150
a_0 (threshold)	+0.35	+0.27168	0	0	300
W_out_eff (readout × gain)	-0.00031521	-8.2305e-05	0.06595	0.4076	1275
readout_bias	+0	+0.0077873	0	0.0891	17
ic.ic (per-variant init)	+0.17587	+0.17587	0.3587	0.3587	1500

srnn-no-adapt-no-dales-skip

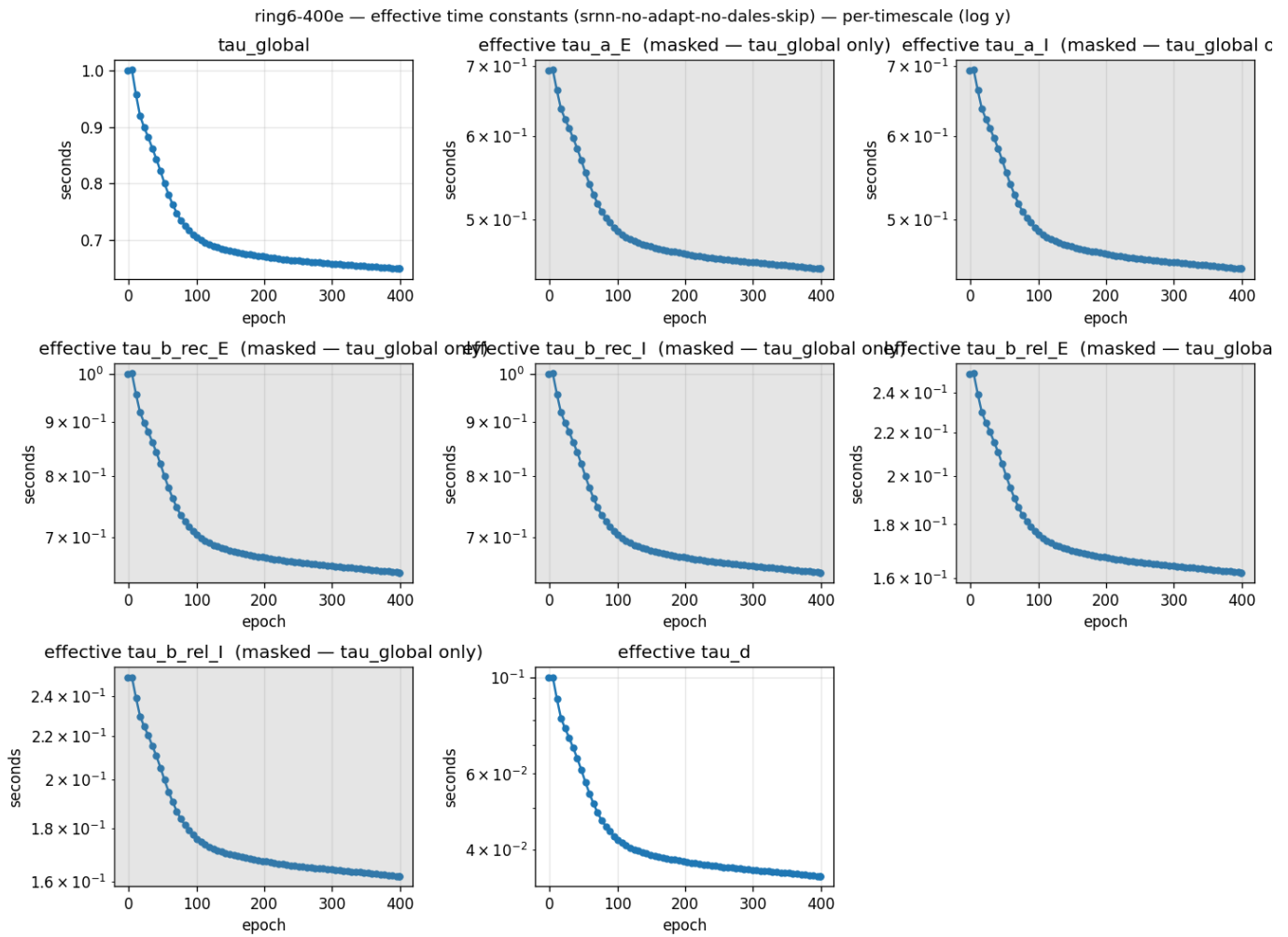
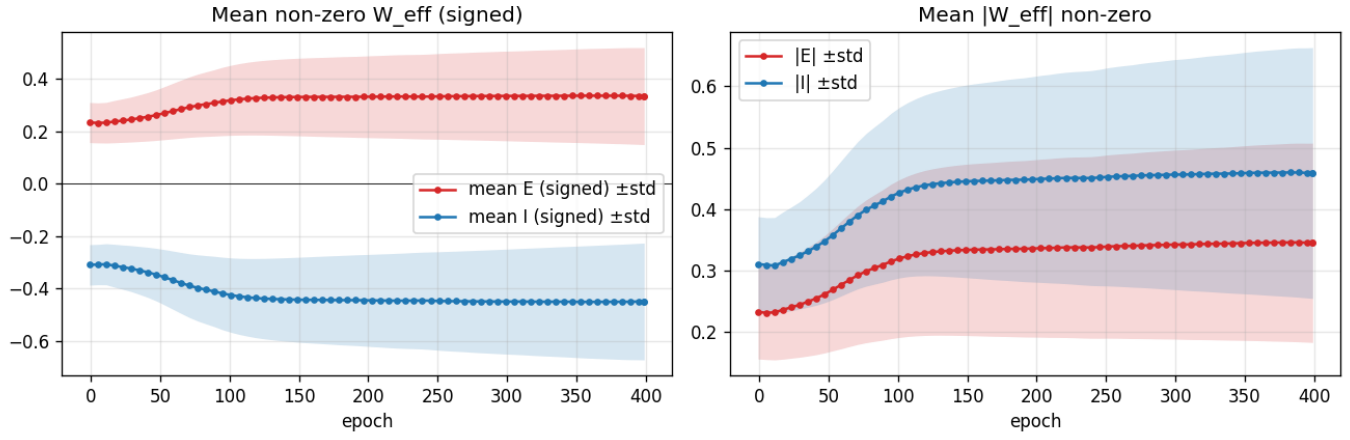
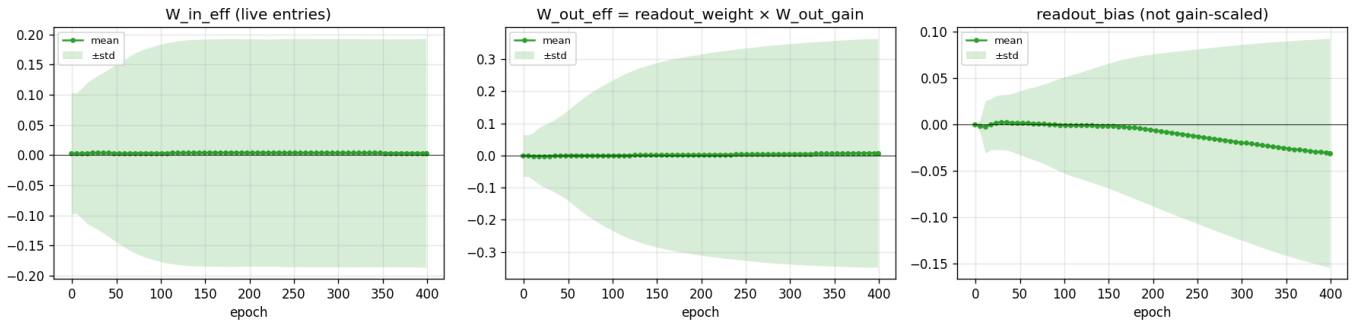


Figure 15: Tau evolution

ring6-400e — recurrent E vs I weights (srnn-no-adapt-no-dales-skip)

Figure 16: W_{EI} evolution

ring6-400e — input / output weights (srnn-no-adapt-no-dales-skip)

Figure 17: W_{io} evolution

offsets and threshold (srnn-no-adapt-no-dales-skip) — scalar-drive

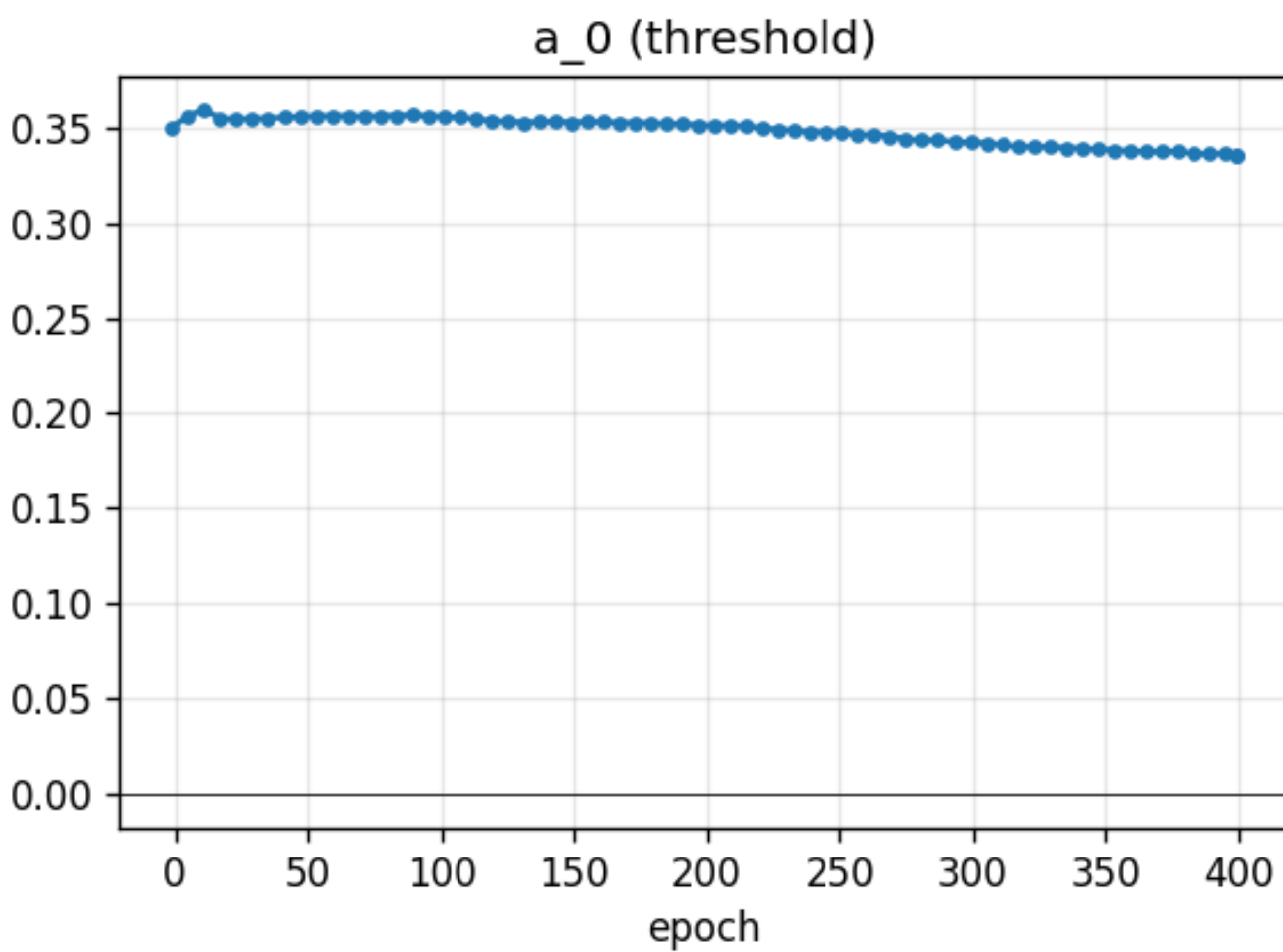


Figure 18: Offsets and threshold

srnn-no-adapt-no-dales-skip — parameter table

```
# ring6-400e - variant srnn-no-adapt-no-dales-skip (k=3)
# Effective values (post-transform). Means/stds computed over the indicated dimension.
```

effective parameter	init mean	final mean	init std	final std	n

tau_global (s)	+1	+0.64821	-	-	scalar
tau_d (s)	+0.1	+0.034793	0	0	300
tau_b_rec_E (s)	+1	+0.64821	0	0	150
tau_b_rel_E (s)	+0.25	+0.16205	0	0	150
tau_b_rec_I (s)	+1	+0.64821	0	0	150
tau_b_rel_I (s)	+0.25	+0.16205	0	0	150
W_eff (E src, signed)	+0.2326	+0.33374	0.07663	0.1849	15048
W_eff (I src, signed)	-0.31062	-0.45023	0.07727	0.2228	14977
W_eff (E src)	+0.23267	+0.34538	0.07641	0.1622	15048
W_eff (I src)	+0.31062	+0.45904	0.07724	0.2041	14977
W_in_eff (input neurons)	+0.0028809	+0.00338	0.1007	0.1897	1275
a_0 (threshold)	+0.35	+0.3357	0	0	300
W_out_eff (readout × gain)	+0.0003819	+0.0081736	0.0651	0.3561	1275
readout_bias	+0	-0.031	0	0.1233	17
ic.ic (per-variant init)	+0.13418	+0.13418	0.4813	0.4813	1500

srnn

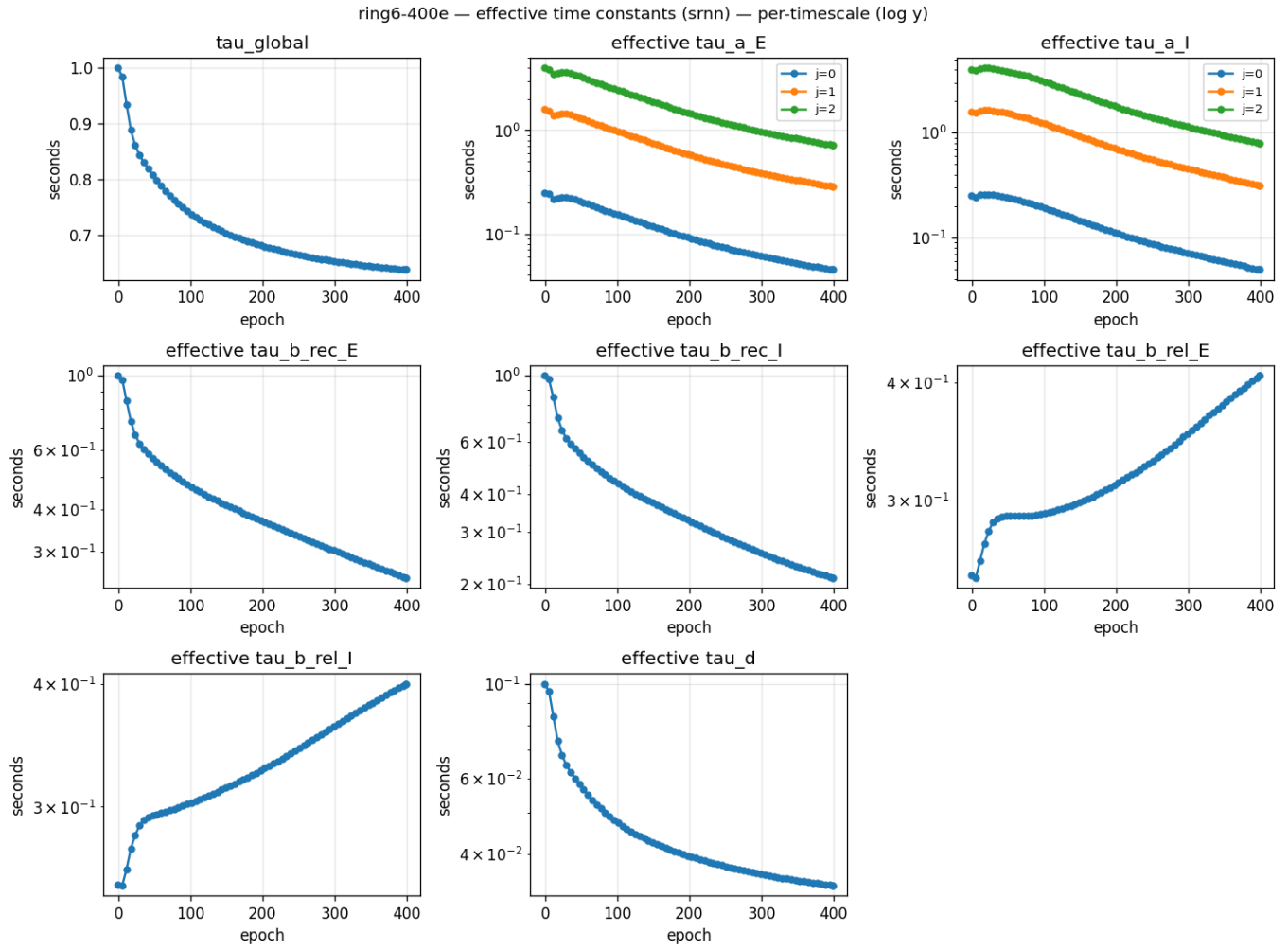
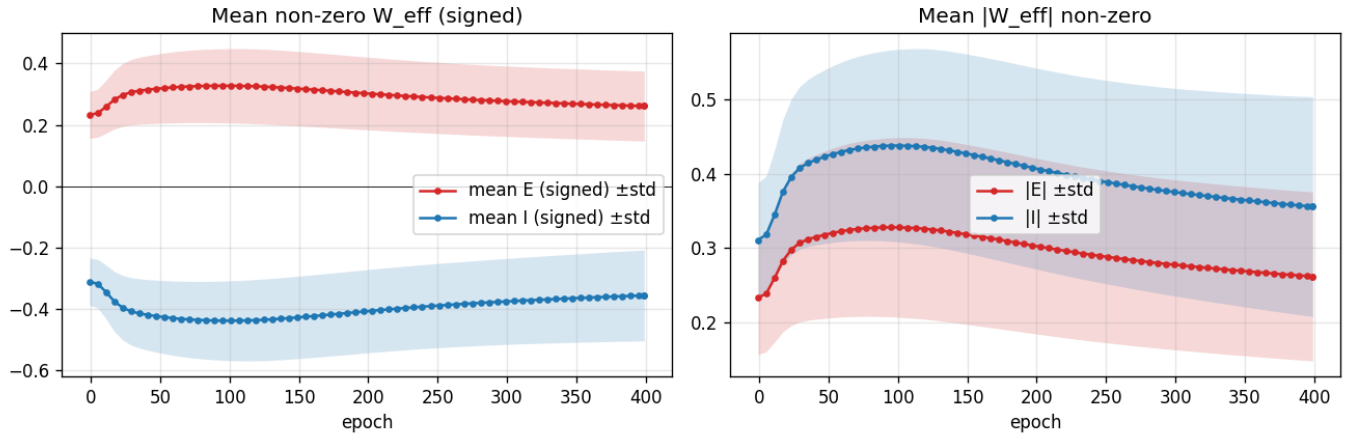
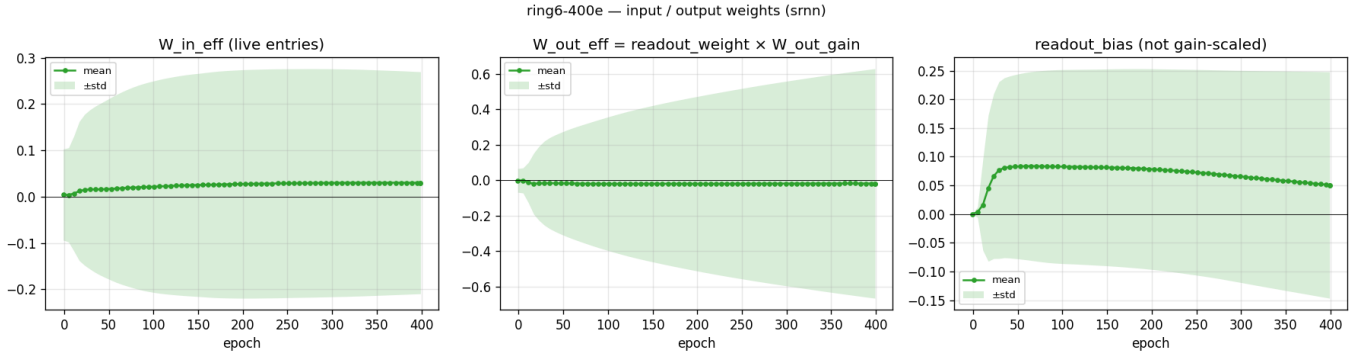


Figure 19: Tau evolution

ring6-400e — recurrent E vs I weights (srnn)

Figure 20: W_{EI} evolutionFigure 21: W_{io} evolution

ring6-400e — offsets and threshold (srnn) — scalar-driven (vec frozen)

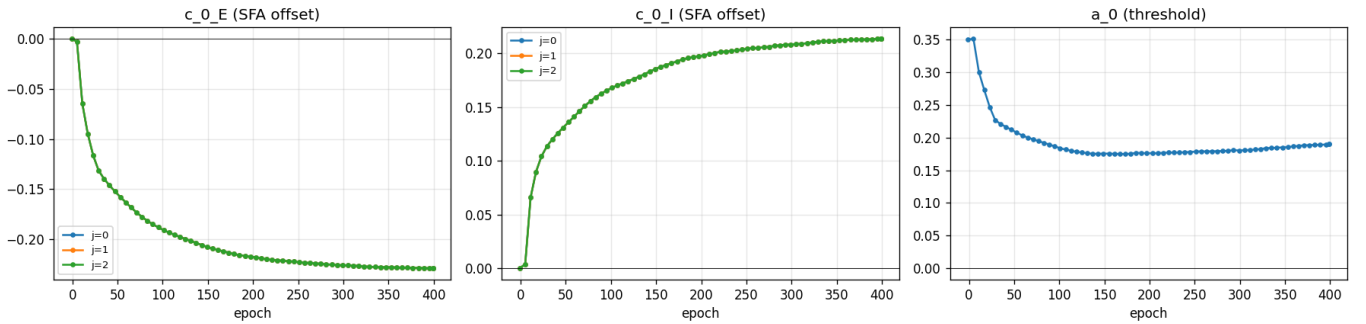


Figure 22: Offsets and threshold

srnn — parameter table

```
# ring6-400e - variant srnn (k=4)
# Effective values (post-transform). Means/stds computed over the indicated dimension.
```

effective parameter	init mean	final mean	init std	final std	n

tau_global (s)	+1	+0.63703	-	-	scalar
tau_d (s)	+0.1	+0.033836	0	0	300
tau_a_E[j=0] (s)	+0.25	+0.044977	0	0	150
tau_a_E[j=1] (s)	+1.5896	+0.28598	0	0	150
tau_a_E[j=2] (s)	+4	+0.71963	0	0	150
tau_a_I[j=0] (s)	+0.25	+0.049557	0	0	150
tau_a_I[j=1] (s)	+1.5896	+0.3151	0	0	150
tau_a_I[j=2] (s)	+4	+0.79291	0	0	150
tau_b_rec_E (s)	+1	+0.25157	0	0	150
tau_b_rel_E (s)	+0.25	+0.40665	0	0	150
tau_b_rec_I (s)	+1	+0.21045	0	0	150
tau_b_rel_I (s)	+0.25	+0.40069	0	0	150
W_eff (E src, signed)	+0.23267	+0.26156	0.07641	0.1139	15048
W_eff (I src, signed)	-0.31062	-0.3557	0.07724	0.1482	14977
W_eff (E src)	+0.23267	+0.26156	0.07641	0.1139	15048
W_eff (I src)	+0.31062	+0.3557	0.07724	0.1482	14977
W_in_eff (input neurons)	+0.004538	+0.030117	0.09841	0.2398	1275
c_E[j=0] (SFA coupling)	+0.05	+0.020863	0	0	150
c_E[j=1] (SFA coupling)	+0.05	+0.020863	0	0	150
c_E[j=2] (SFA coupling)	+0.05	+0.020863	0	0	150
c_0_E[j=0] (SFA offset)	+0	-0.22889	0	0	150
c_0_E[j=1] (SFA offset)	+0	-0.22889	0	0	150
c_0_E[j=2] (SFA offset)	+0	-0.22889	0	0	150
c_I[j=0] (SFA coupling)	+0.05	+0.038609	0	0	150
c_I[j=1] (SFA coupling)	+0.05	+0.038609	0	0	150
c_I[j=2] (SFA coupling)	+0.05	+0.038609	0	0	150
c_0_I[j=0] (SFA offset)	+0	+0.21352	0	0	150
c_0_I[j=1] (SFA offset)	+0	+0.21352	0	0	150
c_0_I[j=2] (SFA offset)	+0	+0.21352	0	0	150
a_0 (threshold)	+0.35	+0.18983	0	0	300
W_out_eff (readout × gain)	-0.001299	-0.018359	0.067	0.648	1275
readout_bias	+0	+0.050641	0	0.1973	17
ic.ic (per-variant init)	+0.1766	+0.1766	0.3557	0.3557	1500

srnn-skip

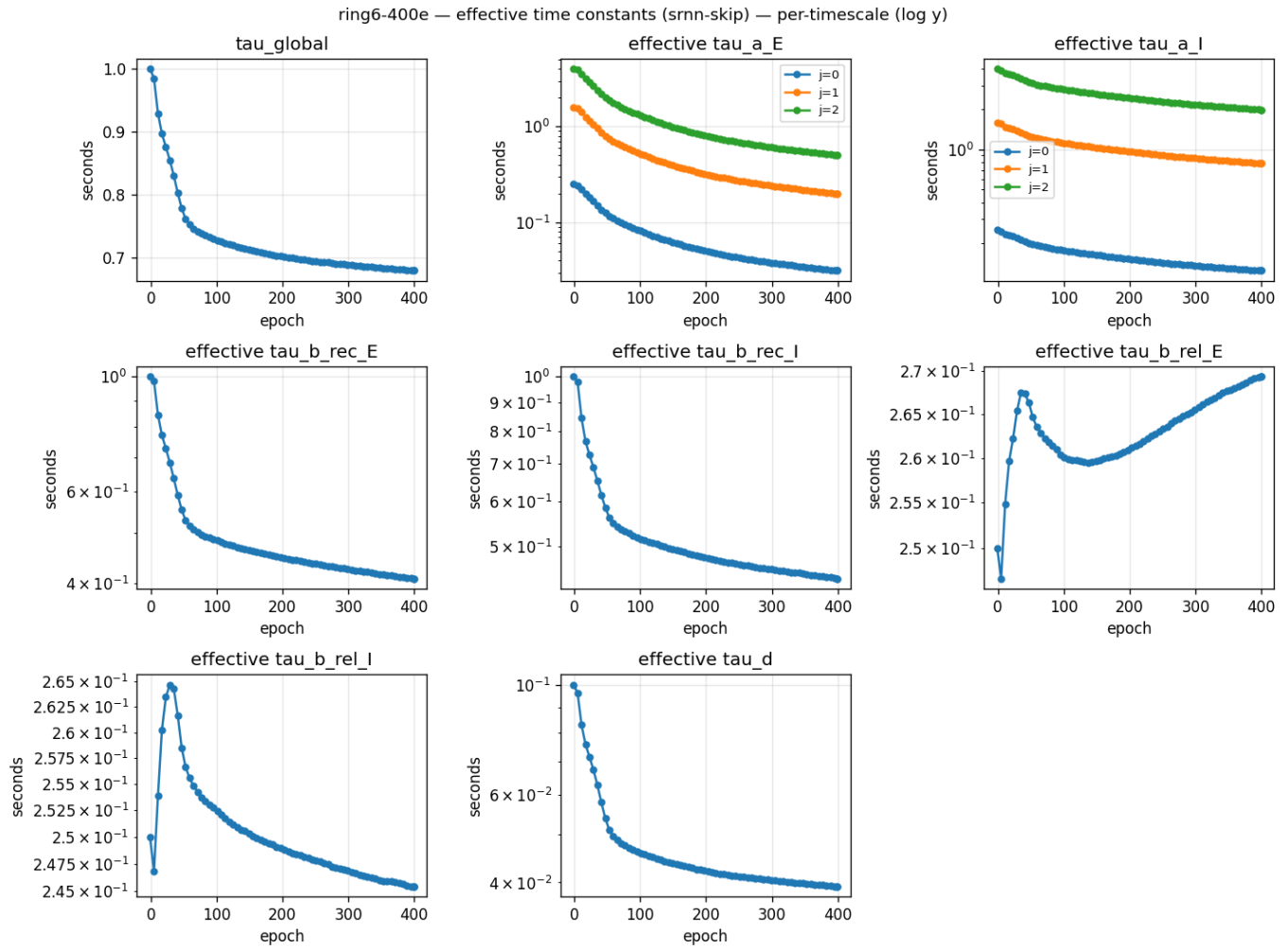
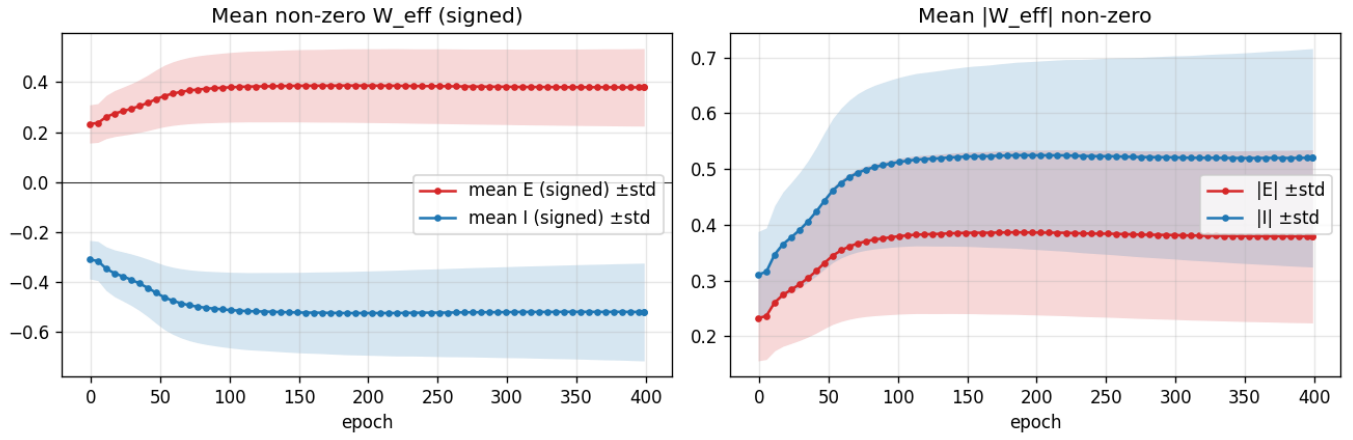
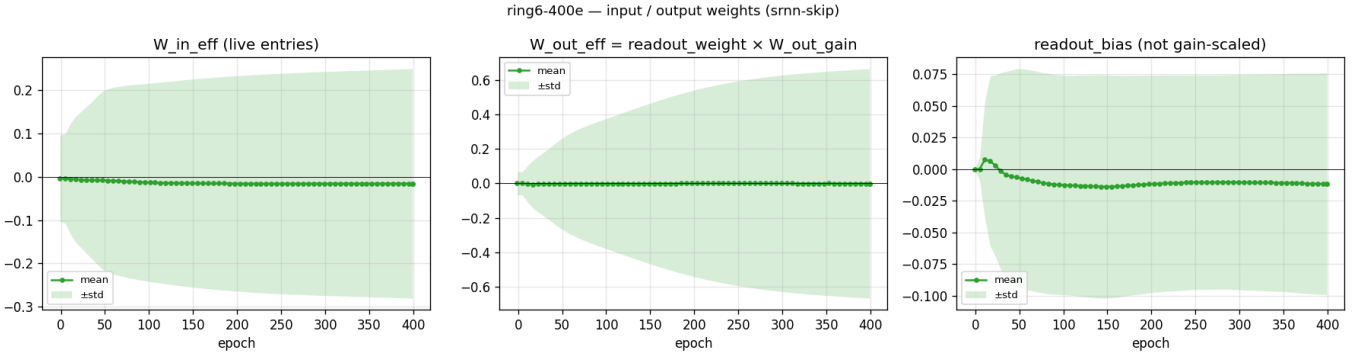


Figure 23: Tau evolution

ring6-400e — recurrent E vs I weights (srnn-skip)

Figure 24: W_{EI} evolutionFigure 25: W_{io} evolution

ring6-400e — offsets and threshold (srnn-skip) — scalar-driven (vec frozen)

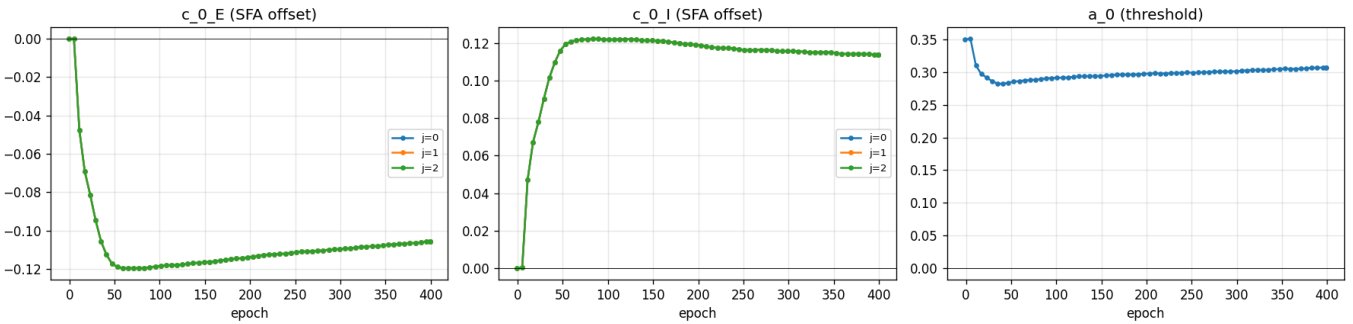


Figure 26: Offsets and threshold

srnn-skip — parameter table

```
# ring6-400e - variant srnn-skip (k=5)
# Effective values (post-transform). Means/stds computed over the indicated dimension.
```

effective parameter	init mean	final mean	init std	final std	n

tau_global (s)	+1	+0.67934	-	-	scalar
tau_d (s)	+0.1	+0.039279	0	0	300
tau_a_E[j=0] (s)	+0.25	+0.0314	0	0	150
tau_a_E[j=1] (s)	+1.5896	+0.19965	0	0	150
tau_a_E[j=2] (s)	+4	+0.50241	0	0	150
tau_a_I[j=0] (s)	+0.25	+0.12391	0	0	150
tau_a_I[j=1] (s)	+1.5896	+0.78789	0	0	150
tau_a_I[j=2] (s)	+4	+1.9826	0	0	150
tau_b_rec_E (s)	+1	+0.40795	0	0	150
tau_b_rel_E (s)	+0.25	+0.26933	0	0	150
tau_b_rec_I (s)	+1	+0.43798	0	0	150
tau_b_rel_I (s)	+0.25	+0.24543	0	0	150
W_eff (E src, signed)	+0.23267	+0.37937	0.07641	0.1551	15048
W_eff (I src, signed)	-0.31062	-0.52007	0.07724	0.1957	14977
W_eff (E src)	+0.23267	+0.37937	0.07641	0.1551	15048
W_eff (I src)	+0.31062	+0.52007	0.07724	0.1957	14977
W_in_eff (input neurons)	-0.0036791	-0.015529	0.09946	0.2645	1275
c_E[j=0] (SFA coupling)	+0.05	+0.051349	0	0	150
c_E[j=1] (SFA coupling)	+0.05	+0.051349	0	0	150
c_E[j=2] (SFA coupling)	+0.05	+0.051349	0	0	150
c_0_E[j=0] (SFA offset)	+0	-0.10566	0	0	150
c_0_E[j=1] (SFA offset)	+0	-0.10566	0	0	150
c_0_E[j=2] (SFA offset)	+0	-0.10566	0	0	150
c_I[j=0] (SFA coupling)	+0.05	+0.053623	0	0	150
c_I[j=1] (SFA coupling)	+0.05	+0.053623	0	0	150
c_I[j=2] (SFA coupling)	+0.05	+0.053623	0	0	150
c_0_I[j=0] (SFA offset)	+0	+0.11367	0	0	150
c_0_I[j=1] (SFA offset)	+0	+0.11367	0	0	150
c_0_I[j=2] (SFA offset)	+0	+0.11367	0	0	150
a_0 (threshold)	+0.35	+0.30654	0	0	300
W_out_eff (readout × gain)	+0.00099875	-0.00040111	0.06582	0.666	1275
readout_bias	+0	-0.011576	0	0.08749	17
ic.ic (per-variant init)	+0.17477	+0.17477	0.3568	0.3568	1500